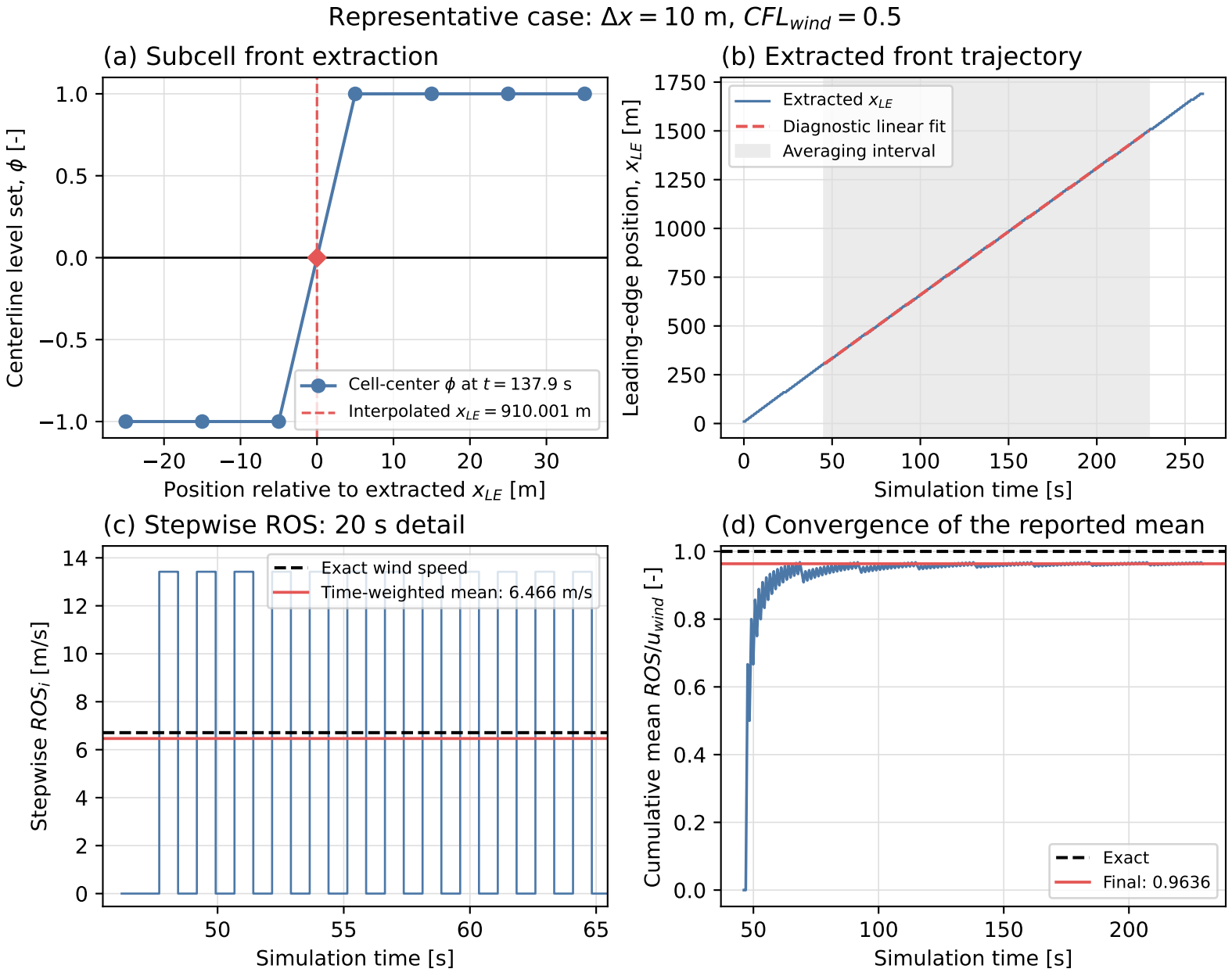
Representative case: $\Delta x = 10$ m, $CFL_{wind} = 0.5$
(a) Subcell front extraction
Centerline level set, ϕ [-]
Position relative to extracted x_{LE} [m]
Cell-center ϕ at $t = 137.9$ s
Interpolated $x_{LE} = 910.001$ m
(b) Extracted front trajectory
Leading-edge position, x_{LE} [m]
Simulation time [s]
Extracted x_{LE}
Diagnostic linear fit
Averaging interval
(c) Stepwise ROS: 20 s detail
Stepwise ROS_i [m/s]
Simulation time [s]
Exact wind speed
Time-weighted mean: 6.466 m/s
(d) Convergence of the reported mean
Cumulative mean ROS/u_{wind} [-]
Simulation time [s]
Exact
Final: 0.9636